Enrolment No./Seat No_____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–I & II EXAMINATION – WINTER 2024

Subject Code:BE01000041 **Date:04-01-2025**

Subject Name:Mathematics-I

Time:10:30 AM TO 01:30 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

Q.1	**(a)**	Find $\lim_{x\to 0}\left(\frac{1}{x^2}-\frac{1}{sin^2x}\right)$	**03**
	(b)	Write second derivative test for local extrema and find the extrema of $f(x) = x^4 - 4x^3 + 10$	**04**
	(c)	Expand e^x into the power series of x . Find number of terms required to calculate e with an error less than 10^{-4} .	**07**
Q.2	**(a)**	Define Beta and Gamma functions and evaluate $\int_0^\infty \frac{x^3(1+x^2)}{(1+x)^{10}}dx$	**03**
	(b)	Discuss the convergence of $\int_{-\infty}^{\infty}\frac{1}{1+x^2}dx$	**04**
	(c)	The circle $x^2 + y^2 = a^2$ is rotated about x −axis to generate a sphere, find its volume and sketch the region.	**07**
		OR	
	(c)	Find the surface area of a solid generated by the revolution of the circle $r = 2acos\theta$ about the initial line.	**07**
Q.3	**(a)**	Determine whether $\lim_{(x,y)\to(0,0)}\frac{x^6-y^2}{x^3y}$ exists, find it if exists.	**03**
	(b)	Applying the chain rule find $\frac{dw}{dt}$ $if\ w = x^2y - y^2$, where $x = sint$ and $y = e^t$ also find $\frac{dw}{dt}$ at $t = 0$	**04**
	(c)	Find the extreme value of the function $f(x,y) = x^3 + y^3 + 3x^2 - 3y^2$	**07**
		OR	
Q.3	**(a)**	If $u = \tan^{-1}(y/x)$ prove that $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$	**03**
	(b)	Find the tangent plane and normal line of the surface $f(x,y,z) = x^2 + y^2 + z - 9 = 0$ at point $(1,2,4)$	**04**
	(c)	Applying Lagrange multipliers to determine the dimension of a rectangular box, open at the top, having volume $32\ ft^3$, and requiring the least amount of material for its construction.	**07**
Q.4	**(a)**	State sandwich theorem for the sequence and discuss the convergence of the sequence $\left\{\frac{\cos n}{n}\right\}$	**03**
	(b)	Test the convergence of the series $\sum_{n=0}^{\infty}\left(\frac{4^n+3}{5^n}\right)$, also find the sum of the series if it is convergent.	**04**
	(c)	Applying appropriate test find the interval of convergence and radius of convergence of the series $\sum_{n=0}^{\infty}\frac{(x-5)^n}{n^2}$	**07**
		OR	

Q.4 **(a)** Investigate the convergence of the series $\sum_{n=0}^{\infty} \frac{4^n n! n!}{(2n!)}$ **03**

(b) Discuss the convergence of the series and find it's sum $\sum_{n=0}^{\infty} \frac{1}{n(n+1)}$ **04**

(c) Applying appropriate test find the interval of convergence and radius of convergence of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{3^n(n+1)}$ **07**

Q.5 **(a)** Evaluate $\iint_R y^2 x \, dA$ over the rectangular region $R = \{(x, y): -3 \leq x \leq 2, \ 0 \leq y \leq 1\}$ **03**

(b) Use double integration to determine the area of the region R inclosed between the parabola $y = \frac{1}{2}x^2$, and the line $y = 2x$, also sketch the region. **04**

(c) Applying change of the order of the integration to evaluate $\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2} \, dx \, dy$, also sketch the region. **07**

OR

Q.5 **(a)** Evaluate $\int_0^1 \int_x^1 \int_0^{y-x} dz dy dx$ **03**

(b) Evaluate the following integral by changing in to the polar co ordinate. **04**
$\iint_R e^{x^2+y^2} dA$ where R is the semi circular region bounded by the x-axis and the curve $y = \sqrt{1 - x^2}$

(c) A thin plate covers the triangular region bounded by the x-axis and the line $x = 1$ and $y = 2x$ in the first quadrant. The plate density at the point (x, y) is $\delta(x, y) = 6x + 6y + 6$ find the plate's mass, first moments and the centre of mass about the coordinate axis. **07**

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Enrolment No./Seat No_____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–I&II EXAMINATION – SUMMER 2025

Subject Code:BE01000041 **Date:05-06-2025**

Subject Name:Mathematics-I

Time:10:30 AM TO 01:30 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	Investigate the convergence of $\int_0^1 \frac{1}{1-x}\,dx$.	**03**
	(b)	Define beta and gamma functions. What is the relationship between beta and gamma functions?	**04**
	(c)	The region between the curve $y=\sqrt{x}, 0 \le x \le 4$ and the x-axis is revolved about the x-axis to generate a solid. Find its volume.	**07**
Q.2	**(a)**	Evaluate $\lim_{x\to 0} \frac{3x-sinx}{x}$ using L'Hospital rule.	**03**
	(b)	Find the Taylor series generated by $f(x)=e^x$ at $x=2$.	**04**
	(c)	Find the Maclaurin series for $f(x)=(1+x)^k$ where k is any real number. Using it, find the Maclaurin series for the function $\frac{1}{1-x}$.	**07**
		OR	
	(c)	Find the local extreme values of the function $f(x)=3x^4-2x^3-6x^2+6x+1$.	**07**
Q.3	**(a)**	Test the the series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ for convergence or divergence.	**03**
	(b)	Test the convergence of the series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n}$.	**04**
	(c)	Define absolutely convergent series and conditionally convergent series. Investigate the convergence of the series $\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$.	**07**
		OR	
Q.3	**(a)**	Investigate the convergence of the series $\sum_{n=1}^{\infty} \frac{2^n}{n!}$.	**03**
	(b)	Find the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.	**04**
	(c)	For what values of x does the series $\sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$ converge?	**07**
Q.4	**(a)**	If $f(x,y)=\frac{xy^2}{x^2+y^4}$, does $\lim_{(x,y)\to(0,0)} f(x,y)$ exist?	**03**
	(b)	If $z=x^2y+3xy^4$, where $x=\sin 2t$ and $y=\cos t$, find $\frac{dz}{dt}$ when $t=0$.	**04**
	(c)	Find the extreme values of the function $f(x,y)=x^2+2y^2$ on the circle $x^2+y^2=1$.	**07**

OR

Q.4	**(a)**	Find the equation of the tangent plane at the point $(-2,1,-3)$ to the ellipsoid $\frac{x^2}{4}+y^2+\frac{z^2}{9}=3$.	**03**
	(b)	Find the directional derivative of the function $f(x,y)=x^2y^3-4y$ at the point $(2,-1)$ in the direction of the vector $\boldsymbol{v}=2\boldsymbol{i}+5\boldsymbol{j}$.	**04**
	(c)	Find the local extreme values of the function $f(x,y)=xy-x^2-y^2-2x-2y+4$.	**07**
Q.5	**(a)**	Evaluate $\int_0^3\int_0^2(4-y^2)dydx$	**03**
	(b)	Evaluate $\iint_R (3x+4y^2)dA$ where R is the region in the upper half-plane bounded by the circles $x^2+y^2=1$ and $x^2+y^2=4$.	**04**
	(c)	Sketch the region of integration, reverse the order of integration and evaluate the integral $\int_0^1\int_x^1 \sin(y^2)\,dydx$.	**07**

OR

Q.5	**(a)**	Evaluate the triple integral $\iiint_B xyz^2\,dV$ where B is the rectangular box given by $B=\{(x,y,z): 0\le x\le 1, -1\le y\le 2, 0\le z\le 3\}$.	**03**
	(b)	Find the area of the region R bounded by $y=2x^2$ and $y^2=4x$.	**04**
	(c)	Find the volume of the solid that lies under the paraboloid $z=x^2+y^2$ and above the region R in the xy-plane bounded by the lines $y=2x$ and the parabola $y=x^2$.	**07**

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Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025

Subject Code: BE01000041/BE01R00041 **Date: 22-12-2025**

Subject Name: Mathematics-I

Time: 02:30 PM TO 05:30 PM **Total Marks: 70**

Instructions

1. Attempt all questions.

2. Make suitable assumptions wherever necessary.

3. Figures to the right indicate full marks.

4. Simple and non-programmable scientific calculators are allowed.

			Marks
Q.1	**(a)**	Define Improper integral of Second kind and test the convergence of the integral $\int_0^1 \frac{5dx}{(x-1)^2}$	**03**
	(b)	Define Gamma function. Prove that $\Gamma(n) = 2\int_0^\infty e^{-x^2} x^{2n-1} dx$	**04**
	(c)	Obtain the Maclaurin's series expansion of $\frac{1}{1-x}$ for $\lvert x\rvert < 1$. Using it, expand $log_e(1-x)$ in powers of x .	**07**
Q.2	**(a)**	Define Beta function. Show that $\beta(m,n) = \beta(n,m)$	**03**
	(b)	Find local extreme value of the function $f(x) = 2x^3 - 6x$	**04**
	(c)	Define geometric series and explain its convergence. Test the convergence for the following series. $(i)\sum_{n=1}^{\infty} \frac{1}{4n^2-1}$ $(ii)\sum_{n=0}^{\infty} 2^{n/2}$	**07**
		OR	
	(c)	Define nth term test for the divergence of a series. Test the convergence for the following series. $(i)\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}}$ $(ii)\sum_{n=1}^{\infty} nsin\left(\frac{1}{n}\right)$	**07**
Q.3	**(a)**	Evaluate : $\lim_{x\to 0}\left(\frac{1}{x} - \frac{1}{\sin x}\right)$	**03**
	(b)	Find the area generated by revolving the curve $y = x,\ x \in [0,1]$	**04**
	(c)	Find radius and interval of convergence of the power series $\sum_{n=0}^{\infty} \frac{nx^n}{2^n}$	**07**
		OR	

(a) Evaluate : $\lim_{x\to 0}\left(\frac{1}{x^2}\right)^{\tan x}$ **03**

(b) Find the volumes of the solids generated by revolving the region under the curve $y=\sqrt{x}$ over the interval [1, 4] about the x-axis. **04**

(c) Define Alternating series. Verify whether the series $\sum_{n=1}^{\infty}\frac{(-1)^n n^2}{n^3+1}$ is conditionally convergent or not. **07**

Q.4 (a) Evaluate the limit : $\lim_{(x,y)\to(0,0)}\frac{\sqrt{x}-\sqrt{y}}{\sqrt{x}+\sqrt{y}}$ **03**

(b) Find Tangent plane and Normal line to the surface $xy^2+2yz^2+3zx^2=6$ at the point (1,1,1) . **04**

(c) Find the shortest and the longest distances of the point (1,1,1) from the sphere $x^2+y^2+z^2=1$ **07**

OR

(a) Discuss the continuity of $f(x,y)$ at the origin. where, **03**

$$f(x,y)=\begin{cases}\frac{2x^2y}{x^4+2y^2} & (x,y)\neq(0,0)\\ 0 & (x,y)=(0,0)\end{cases}$$

(b) Find the directional derivative of $f(x,y)=x^2e^{xy}$ at the point (1, 0) in the direction of $\bar{u}=\hat{i}-\hat{j}$ **04**

(c) Find critical points of the function $f(x,y)=x^3-y^3-3x+3y+1$. Also find local maxima, local minima and saddle points, if any. **07**

Q.5 (a) Evaluate : $\iint_R (x+y)^2\,dA\ ;\ R=[0,1]\times[0,2]$ **03**

(b) Evaluate the integral $\int\int r\sqrt{a^2-r^2}drd\theta$ over the region of the upper half of the circle $r=a\cos\theta$ **04**

(c) Evaluate the integral $\int_0^{\infty}\int_x^{\infty}e^{-y^2}dydx$ by changing the order of the integral. **07**

OR

(a) Evaluate $\iint_R dA$ where R is the region bounded by the parabolas $y = x^2$ and $x = y^2$. **03**

(b) Evaluate : $\int_0^1 \int_0^1 \int_{\sqrt{x^2+y^2}}^2 z \, dzdydx$ **04**

(c) Find the volume of the region bounded by a paraboloid $z = 1 - x^2 - y^2$, a plane $z = 1$ and XY-plane. **07**

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